CURRICULUM VITAE

Tim Lichtenberg

Assistant Professor | Kapteyn Astronomical Institute | University of Groningen

tim.lichtenberg@rug.nl formingworlds.space rug.nl/staff/tim.lichtenberg

I study the physical and chemical processes that drive the growth and long-term evolution of rocky planets and exoplanets, from formation through billions of years of geodynamic and climate evolution. My research traces how radioactive heating and planetesimal differentiation set the initial volatile budgets, how magma ocean epochs forge the first atmospheres, and how mantle dynamics and volatile cycling shape planetary environments over geological time. By bridging planetary physics, geochemistry, and exoplanet observations, my research group works to understand the conditions for the origins of life and its cosmic prevalence.

Research Positions

Tenure Track Assistant Professor & Branco Weiss Fellow (University of Groningen, NL) 10/2022–

Simons Collaboration on the Origins of Life Postdoctoral Fellow (University of Oxford, UK) 10/19–09/22

Swiss National Science Foundation Early.Postdoc Mobility Fellow (University of Oxford, UK) 08/18–09/19

Education

Dr. sc. ETH Zurich (Department of Earth Sciences, ETH Zurich, Switzerland) 09/14–07/18

Thesis: Thermal Evolution of Forming Planets: Isotope Enrichment, Differentiation & Volatile Retention

Master of Science Physics (Georg-August-Universität Göttingen, Germany) 09/12–08/14

Thesis: Modeling gravitational instabilities in compact and massive protoplanetary disks with adaptive mesh refinement techniques (grade: 1.0, the best German grade)

Semester abroad (Peking University, People's Republic of China) 09/12–01/13

Studies of Chinese language and culture

Bachelor of Science Physics (Georg-August-Universität Göttingen, Germany) 09/09–08/12

Thesis: Constraining Exoplanet Characteristics with Asteroseismology (grade: 1.0, the best German grade)

Honours & Awards

Europlanet Early-Career Medal, Europlanet Society 09/25

Planetary and Solar System Sciences Division Outstanding Early Career Scientist Award, European Geosciences Union (EGU) 04/23

Winton Award for Early Achievement in Geophysics, Royal Astronomical Society 01/22

Doctoral Thesis Award 2019, German Astronomical Society 09/19

PhD Prize for Planetary Systems and Astrobiology 2018, International Astronomical Union 06/19

European Astrobiology Network Association, Horneck-Brack Presentation Award, 2nd prize 09/18

'Dr. Berliner - Dr. Ungewitter' award for M.Sc. thesis, Georg-August-Universität Göttingen 12/14

Master's diploma awarded 'with distinction', Georg-August-Universität Göttingen 08/14

Grants & Fellowships

18 grants and fellowships in total, incl. 13 as PI (>€3M), 4 as Co-I

Selected list. Full list online

Open eScience Project *inversePROTEUS* (€531'000, incl. about 3 person-years of research software engineering, *Scalable Physics-based Inference of Exoplanet Evolution*), Netherlands eScience Center, PI 10/26

Computing Time on National Computer Facilities, Small Compute grant EINF-19723 (535'000 SBU on the Snellius supercomputer, 50'000 CPU hours on the SURF Research Cloud, *Pilot for the Interra Project exoplanet simulation ensemble on Snellius*), Dutch Research Council (NWO) & SURF, PI 09/26

AEThER Collaboration, Sloan Foundation, Co-I 07/25

European Research Council's (ERC) Starting Grant MagmaWorlds (€1.5M), PI 09/25

Dutch Research Council (NWO) Consortium Grant (€329'065 out of €6.7M in total, *PRELIFE - Pathways, Reactions, and Environments leading to LIFE*), work package leader, Co-I 11/24

NOVA Consortium Grant (LIFE: Mid-infrared detection of biosignatures in the atmospheres of terrestrial exoplanets), PI: van der Tak (fte + cash € 249'000), Co-I 07/24

Open eScience Early Career Project PROTEUS (€134'000, *An Adaptive Numerical Framework to Simulate Lava Exoplanets*), Netherlands eScience Center, PI 12/23

Faculty Research Grant, Faculty of Science and Engineering (€150'000), University of Groningen, PI 12/22

Research Fellowship of the Branco Weiss Foundation (*CHF 500'000, Molten Exoplanets as a Window into the Earliest Earth*), PI 08/22

Non-stipendiary Junior Research Fellowship, St Cross College, University of Oxford, PI 06/19

Simons Collaboration on the Origins of Life Postdoctoral Fellowship ($276'000, *Geodynamic Regimes during Planetogenesis*), Simons Foundation, PI 12/18

Early Postdoc.Mobility fellowship (*CHF 77'650, Linking Interior-Atmosphere Regimes to Planetary Accretion*), Swiss National Science Foundation, PI 12/17

Conference and Institute Seminar Presentations

174 presentations in total, incl. 149 oral presentations and 25 posters; 86 invited ‡

Selected list: invited talks since 2021. Full list online

CONFERENCES AND WORKSHOPS

Layers of Understanding: Model Intercomparisons of Exoplanet Interiors, Heidelberg, DE ‡ PDF 04/26

International Conference on Exoplanets and Planet Formation, Shanghai, CN ‡ PDF 12/25

Europlanet Science Congress, Helsinki, FI ‡ PDF 09/25

Origins Federation Conference, Cambridge, UK ‡ PDF 09/24

The Geoscience of Exoplanets: Going Beyond Habitability, International Space Science Institute, Bern, CH ‡ PDF 04/24

Exoclimes VI, Exeter, UK ‡ PDF 06/23

Gordon Research Conference – Origins of Solar Systems, South Hadley, US ‡ PDF 06/23

European Geosciences Union (EGU) General Assembly, Vienna, AUT ‡ PDF 04/23

Protostars and Planets VII, Kyoto, JP ‡ PDF 04/23

International Astronomical Union General Assembly, Division F: Planetary Systems and Astrobiology, Busan, KR ‡ PDF 08/22

Annual Meeting of the European Astronomical Society, Valencia, ESP ‡ PDF 06/22

Goldschmidt 21, Chemical geodynamics throughout the Solar System, Lyon, FR (virtual) ‡ PDF 07/21

Habitable Worlds 21, NASA NExSS, NExScI, US (virtual) ‡ PDF 02/21

COLLOQUIA AND SEMINAR TALKS

Institute of Astronomy, KU Leuven, Leuven, BE ‡ PDF 10/26

Globe Institute, Centre for Star and Planet Formation, Copenhagen, DK ‡ PDF 02/26

Max Planck Institute for Solar System Research, Göttingen, DE ‡ PDF 05/24

Royal Astronomical Society Ordinary Meeting, London, UK ‡ PDF 10/23

Harvard University, Department of Earth and Planetary Sciences, Cambridge, US ‡ PDF 06/23

California Institute of Technology, Division of Geological and Planetary Sciences, Yung Lunch Seminar, Pasadena, USA ‡ PDF 07/22

AEThER Collaboration Seminar, Carnegie Earth and Planets Laboratory, Washington DC, USA ‡ PDF 06/22

Imperial College, Department of Earth Sciences, London, UK ‡ PDF 03/22

Heidelberg Initiative for the Origins of Life Colloquium, Max Planck Institute for Astronomy, Heidelberg, DE ‡ PDF 11/21

Lunar and Planetary Institute, USRA, Houston, US (virtual) ‡ PDF 03/21

Postdoc Mentoring

4 mentored postdocs in total, incl. 3 with personal fellowships (2 SNSF, 1 JSPS), 1 ERC-funded

Patrick Bos, Senior Research Software Engineer (ERC Starting Grant *MagmaWorlds*), Kapteyn Institute 03/26–

Mara Attia, SNSF Postdoctoral Fellow, Kapteyn Institute 05/25–

Tadahiro Kimura, JSPS Postdoctoral Fellow, Kapteyn Institute 06/24–03/26

Rob Spaargaren, SNSF Postdoctoral Fellow, Kapteyn Institute 12/23–11/25

Student Mentoring

44 supervised students in total, incl. 8 PhD, 10 MSc, 16 BSc theses, 10 projects/internships

Selected list: PhD students by name, other students summarised. Full list online

PHD STUDENTS

Saphira Brandenberger, exoplanet core evolution, Kapteyn Institute 09/26–

Emeline Decocq, prebiotic climate and ocean levels of the Hadean Earth, Kapteyn Institute 09/25–

Imre Kisvárdai, interior-atmosphere evolution on low-mass exoplanets, Kapteyn Institute 09/25–

Emma Postolec, evolution of magma ocean exoplanet atmospheres, Kapteyn Institute 10/23–

Mariana V. Sastre, evolution of magma ocean exoplanet interiors, Kapteyn Institute 09/23–

Harrison Nicholls, rocky planet evolution (co-supervised with Raymond Pierrehumbert), Oxford Atmospheric Physics 04/23–01/26

Robert J. Graham, condensable-rich exoplanet atmospheres (co-supervised with Raymond Pierrehumbert), Oxford Atmospheric Physics 08/20–09/22

Ryan Boukrouche, the post-runaway greenhouse state on rocky exoplanets (co-supervised with Raymond Pierrehumbert), Oxford Atmospheric Physics 01/20–09/22

THESES AND PROJECTS BY INSTITUTION

Kapteyn Institute and Physics department, University of Groningen: 8 MSc and 15 BSc theses and 5 research projects (partly co-supervised with Mara Attia, Quentin Changeat and Joost van den Born) on magma oceans and lava planets, atmospheric escape, oceans on sub-Neptunes and Earth-like planets, the Hadean Earth, interior structure, steam and rock-vapour atmospheres, and observations with LIFE and Ariel; the projects include the visiting PhD student Anna Grace Ulses (University of Washington) and a PhD research project with Beat Hubmann 04/23–

St Cross College, University of Oxford: pastoral support and mentoring for 16 MSc and DPhil students 10/19–09/22

University of Bern: PhD research projects with Tobias G. Meier (co-supervised with Dan J. Bower) on the interior dynamics of tidally locked super-Earths 06/20–09/22

ETH Zurich: 2 MSc theses, 1 BSc thesis and 3 semester projects (co-supervised with Taras Gerya and Sascha Quanz) on planetesimal hydrology, core formation and pebble accretion, proto-planetary collisions, and the surfaces of airless exoplanets 08/16–10/22

Lectures and Courses

17 courses in total, incl. 11 as lecturer, 6 as teaching/lab assistant

Selected list: lecturer roles since 2021. Full list online

Planetary Systems, BSc course, Kapteyn Institute, lecturer (∼40 students) 09/26–11/26

Exoplanets, MSc course, Kapteyn Institute, lecturer (∼20 students) 02/26–04/26

Planetary Systems, BSc course, Kapteyn Institute, lecturer (∼50 students) 09/25–11/25

Exoplanets, MSc course, Kapteyn Institute, lecturer (∼20 students) 10/24–02/25

Star and Planet Formation, MSc course, Kapteyn Institute, co-lecturer (∼30 students) 02/25–04/25

Exoplanets, MSc course, Kapteyn Institute, lecturer (∼20 students) 10/23–02/24

Hands-on Numerical Astrophysics School for Planet Formation Sciences, summer school, DFG priority programme 1992 “Exoplanet Diversity”, Rauenberg, lecturer (∼30 students) 08/23

Star and Planet Formation, MSc course, Kapteyn Institute, co-lecturer (∼30 students) 02/23–04/23

External/Co-Examination & Defence Committees

24 examinations in total, incl. 7 PhD, 2 MSc, 15 BSc

Selected list: PhD defences by name, other theses summarised. Full list online

PHD DEFENCES

Aditya Arabhavi, planet-forming materials around young stars with JWST, University of Groningen 09/25

Mark Oosterloo, CHNOS in the early stages of planet formation, University of Groningen 01/25

Francisco Ardévol Martinez, machine learning for exoplanet characterisation, University of Groningen 10/24

Keavin Moore, water on terrestrial planets around M dwarfs, McGill University 12/23

Bayron Armando Portilla Revelo, radiative transfer models of planet-forming disks, University of Groningen 12/23

Yapeng Zhang, isotopes and the characterization of extrasolar planets, Leiden Observatory 06/23

Mantas Zilinskas, atmospheres of lava worlds, Leiden Observatory 05/23

MSC AND BSC THESES

University of Groningen: 2 MSc and 15 BSc theses on exoplanet atmospheres and JWST data, the radius valley, the climate of Earth-like exoplanets, protoplanetary disk chemistry, non-LTE line diagnostics, and exoplanet host stars 06/23–07/26

Refereeing & Panel Service

AGU Advances, Astronomical Journal, Astronomy & Astrophysics, Astrophysical Journal, Astrophysical Journal Letters, Earth and Planetary Science Letters, Elements, Icarus, Journal of Geophysical Research: Planets, Monthly Notices of the Royal Astronomical Society, Nature, Nature Astronomy, Nature Communications, Planetary Science Journal, Proceedings of the National Academy of Sciences, Publications of the Astronomical Society of Australia, Science, Science Advances

Belgian Research Foundation (FWO), Cambridge University College Fellowships, Dutch Research Council (NWO), European Research Council (ERC), Max Planck Society (MPG), NASA Emerging Worlds program, NASA Hubble Fellowship Program (NHFP), Natural Sciences and Engineering Research Council of Canada (NSERC), Swiss National Science Foundation (SNSF), UK Research and Innovation (UKRI), US National Science Foundation (NSF)

Astrobiology Science Officer, EGU Planetary and Solar System Sciences Division 06/23–05/25

National & International Collaborations

4 observing programmes with 268.3 hr of telescope time, incl. 1 as PI; member of 9 consortia and research programmes

TELESCOPE TIME

INT/HARPS3 Semester 26B, SI2026b03: *High-Precision RVs for an Expanded Census of Stellar and Planetary Compositions* (62.2 hr), PI 07/26–

INT/HARPS3 Semester 26B, SI2026b01: *HARPS3-NEAR: A Radial-Velocity Census of the Nearest Sun-like Stars* (61.5 hr), PI: Snellen, Co-I 07/26–

JWST Cycle 4, GO 8864: *Surveying Hellish Worlds: Lava Planets as Time Capsules of Thermal Evolution* (100.4 hr), PI: Dang, Co-I 04/25–

JWST Cycle 2, GO 3860: *Phase Curve Observations of TOI-561 b To Study Atmosphere-Interior Exchange* (44.2 hr), PI: Teske, Co-I 02/23–

CONSORTIA & RESEARCH PROGRAMMES

Dutch Research Council (NWO) Consortium PRELIFE – Pathways, Reactions, and Environments leading to LIFE, Co-I 11/24–

ESA PLATO Consortium, member of Exoplanet Science WPs *Atmospheres of PLATO Terrestrial Planets* (WP116400) and *Climate/Atmospheres of PLATO Habitable Zone Planets* (WP116530); Core Science Team member since 2025 12/21–

Co-I (since 2024, previously associate member) of Carnegie-led research program *AEThER: Atmospheric Empirical, Theoretical, and Experimental Research* 09/21–

Associate member of NASA research program Alien Earths: Searching for Habitable Worlds in Nearby Planetary Systems 03/21–

Co-lead of the *Science Team* of the LIFE Collaboration 02/21–01/26

Member of the *Quantifying Habitability* working group of NASA's Nexus for Exoplanet System Science 09/20–

ESA Ariel Consortium, member of the *Exoplanetary Interiors* working group 04/20–

Postdoctoral Fellow of the Simons Collaboration on the Origins of Life (SCOL project) 10/19–09/22

Member of the Large Interferometer for Exoplanets (LIFE) Collaboration 01/19–

Conference & Workshop Organisation

30 events in total, incl. 8 as main organiser

Selected list: events since 2021. Full list online

Rocky Worlds Conference Series (main organiser) Chair 10/22–

Rocky Worlds 5, Berlin, DE SOC 04/28

Lorentz Center Workshop *Characterizing Habitable Exoplanets With Interdisciplinary Expertise*, Leiden, NL SOC 04/26

Rocky Worlds IV (main organiser), Groningen, NL Chair 01/26

Science and Technology for the Era of LIFE, Barcelona, ES SOC 11/25

Lorentz Center Workshop *Atmospheric and interior evolution of planetary magma oceans* (main organiser), Leiden, NL Chair 10/25

Exoclimes VII, Montréal, CA SOC 07/25

Exoplanets V, Leiden, NL Convenor 06/24

Rocky Worlds III, ETH Zurich, CH SOC 01/24

Annual LIFE Science Team Workshop, Berlin, DE Co-Chair 09/23

Goldschmidt 2023, *Unraveling planetary interiors with extraterrestrial samples and space data*, Lyon, FR Convenor 07/23

Origins of Solar Systems Gordon Research Seminar (main organiser), South Hadley, US Co-Chair 06/23

The (geo)chemistry of terrestrial planet formation, Groningen, NL SOC & LOC 03/23

Rocky Worlds Discussions, monthly virtual seminar series (main organiser) SOC 10/22–09/24

Rocky Worlds II (main organiser), University of Oxford, UK Chair 07/22

AGU 2021 session *Multidisciplinary perspectives on the formation and early evolution of terrestrial worlds*, New Orleans, US Convenor 12/21

ESA PLATO Exoplanet Atmospheres Workshop, Berlin, DE SOC 12/21

Lorentz Center Workshop *LIFE – Large Interferometer for Exoplanets* (main organiser), Leiden, NL SOC 06/21

Advanced Teaching & Research Training Courses

14 courses in total, incl. 10 on teaching, education & supervision and 4 research schools

Selected list: research schools summarised. Full list online

TEACHING, EDUCATION & SUPERVISION

Academic Leadership, University of Groningen, NL 09/25

University Teaching Qualification (UTQ), University of Groningen, NL 02/24

Active Bystander Training, Kapteyn Institute, NL 04/23

Coaching PhD students, University of Groningen, NL 01/23

Inclusive Leadership: The Power of Workplace Diversity, University of Colorado, US (online) 06/21

RESEARCH SCHOOLS

4 international research schools on star and planet formation, Solar System chronology and planetary geodynamics: 45th Saas-Fee Advanced Course (Les Diablerets, CH), Les Houches (FR), TRR 170 winter school (Hodenhagen, DE), ISSAC 2013 (UC Santa Cruz, US) 08/13–02/17

Outreach & Public Engagement

50+ outreach activities, incl. 14 media interviews & press features, 4 public talks & 2 panels, 16 blog posts & videos, 22 peer-reviewed Astrobites articles

Selected list: media interviews and press since 2021. Full list online

MEDIA INTERVIEWS & PRESS

Media interview, *Astronomen ontdekken de eerste rotsachtige exoplaneet met een atmosfeer*, KIJK Magazine	07/26
Press coverage for Nicholls et al. Nature Astronomy 2026 on the molten sulfur world L 98-59 d, primary feature *Molten Sulfurous World Blurs Exoplanet Categories* (Sky & Telescope), also covered in ABC News, Space.com, University of Groningen news, and others	03/26
Media interview, *Stormy planets and an unexpected atmosphere*, FSE Science Newsroom, University of Groningen	12/25
Media interviews for NASA, Space.com, phys.org, ScienceDaily, and others on TOI-561 b atmosphere detection	11–12/25
Interview, *Quickfire Questions with Tim Lichtenberg*, Europlanet Magazine	09/25
Media interview, *This Blazing Exoplanet Breaks All the Rules about Alien Atmospheres*, Scientific American	09/25
Media interview, *Exoplanet plate tectonics: A new frontier in the hunt for alien life*, New Scientist	12/24
Media interview, podcast *Die knifflige Suche nach außerirdischem Leben*, Deutschlandfunk	12/24
Media interview, *What is a presumed sign of life doing on a dead comet?*, Science News	04/24
Biographical interview, *Welt der Physik*, Year of Science 2023, German Ministry for Science and Education	04/23
Media interview, *Planeet en ster groeien samen op*, New Scientist Netherlands	12/22
Press coverage for Bonsor+2023 *Nature Astronomy*, University of Groningen, ScienceDaily, and others	11/22
Press coverage for Lichtenberg+2021 *Science*, phys.org, ScienceDaily, UPI, SciTechDaily, and 36+ outlets	01/21

Academic Associations

Royal Netherlands Astronomical Society	05/24–
European Association of Geochemistry	07/21–
International Astronomical Union	05/21–
American Geophysical Union	01/20–
European Astronomical Society	06/19–
Royal Astronomical Society	06/19–
Europlanet Society	03/19–
European Geosciences Union	01/19–
Swiss Society for Astrophysics and Astronomy	01/15–01/24
German Astronomical Society	11/13–
German Physical Society	11/10–